

Labwork 2: Get to know your GPU

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1 Objective

The goal of this labwork is to use `numba.cuda` to get basic information about the GPU: its name, its number of multiprocessors and cores, and its memory size.

2 Environment

My laptop only has an Intel Iris Xe graphics card, and CUDA only works with NVIDIA GPUs. So I ran the code on Google Colab (Linux), which gives access to an NVIDIA Tesla T4 GPU.

3 Method

First, `cuda.detect()` lists the available GPUs. Then `cuda.select_device(0)` selects the first GPU and returns a device object. From this object I read the `id` and `name` attributes, the number of streaming multiprocessors (SM) with `MULTIPROCESSOR_COUNT`, and the compute capability with `compute_capability`. For the memory, `cuda.current_context().get_memory_info()` returns the free and total memory in bytes, so I divide by 1024^3 to get the value in GB.

Numba does not give the number of cores directly. A GPU is divided into SMs, and each SM contains a fixed number of cores. This number depends on the GPU architecture, which is given by the compute capability. Compute capability 7.5 corresponds to the Turing architecture, which has 64 CUDA cores per SM (NVIDIA documentation). So I computed:

$$\text{total cores} = \text{number of SM} \times \text{cores per SM} = 40 \times 64 = 2560$$

This matches the official Tesla T4 datasheet, which gives 2560 CUDA cores. The value 64 is only valid for this GPU: on a GPU with another compute capability, it has to be changed.

```
from numba import cuda

cuda.detect()

gpu = cuda.select_device(0)
print(gpu.id)
print(gpu.name)

sm_count = gpu.MULTIPROCESSOR_COUNT
print(sm_count)

cc = gpu.compute_capability
print(cc)

# Tesla T4 : compute capability 7.5 = Turing architecture = 64 cores per SM
cores_per_sm = 64
```

```

cores = sm_count * cores_per_sm
print(cores)

free, total = cuda.current_context().get_memory_info()
print(total / 1024**3)
print(free / 1024**3)

```

4 Results

Property	Value
Device ID	0
Device name	Tesla T4
Compute capability	7.5
Multiprocessors (SM)	40
Cores per SM	64
Total cores	2560
Total memory	14.56 GB
Free memory	14.46 GB

Table 1: Information about the GPU

Output of `cuda.detect()`:

```

Found 1 CUDA devices
id 0          Tesla T4          [SUPPORTED]
      Compute Capability: 7.5
      PCI Device ID: 4
      PCI Bus ID: 0
      UUID: GPU-60411518-bba0-87d6-6592-1682
      f007169f
      Watchdog: Disabled
      FP32/FP64 Performance Ratio: 32
Summary:
      1/1 devices are supported

```

5 Conclusion

The Tesla T4 has 40 multiprocessors, 2560 cores and about 14.5 GB of memory. This is many more cores than a normal CPU, which is why a GPU can be much faster for tasks that can be split into many small independent parts, like processing all the pixels of an image.